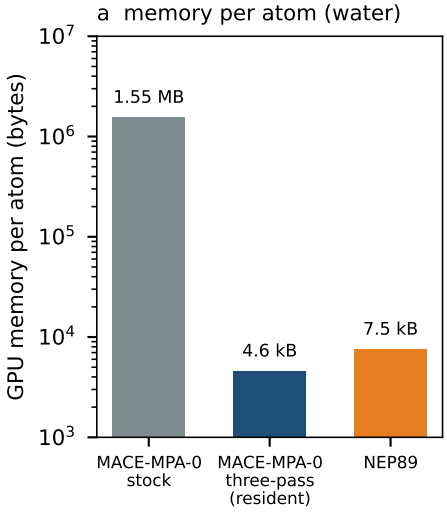
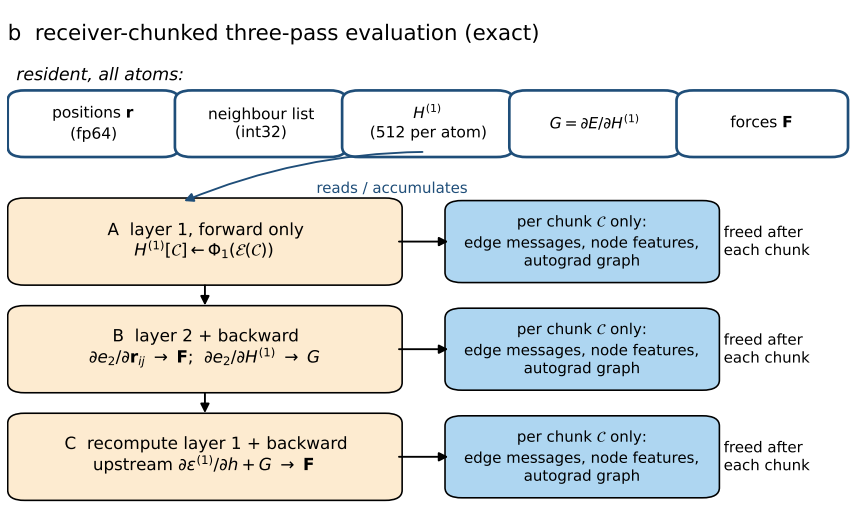

a memory per atom (water)
GPU memory per atom (bytes)
10^7
10^6
10^5
10^4
10^3
1.55 MB
4.6 kB
7.5 kB
MACE-MPA-0 stock
MACE-MPA-0 three-pass (resident)
NEP89


b receiver-chunked three-pass evaluation (exact)
resident, all atoms:
positions r (fp64)
neighbour list (int32)
$H^{(1)}$ (512 per atom)
$G = \partial E/\partial H^{(1)}$
forces F
reads / accumulates
A layer 1, forward only
$H^{(1)}[\mathcal{C}] \leftarrow \Phi_1(\mathcal{E}(\mathcal{C}))$
per chunk $\mathcal{C}$ only: edge messages, node features, autograd graph
freed after each chunk
B layer 2 + backward
$\partial e_2/\partial \mathbf{r}_{ij} \rightarrow \mathbf{F}$; $\partial e_2/\partial H^{(1)} \rightarrow G$
per chunk $\mathcal{C}$ only: edge messages, node features, autograd graph
freed after each chunk
C recompute layer 1 + backward
upstream $\partial \varepsilon^{(1)}/\partial h + G \rightarrow \mathbf{F}$
per chunk $\mathcal{C}$ only: edge messages, node features, autograd graph
freed after each chunk